Model Evaluations

Outline

- Model Evaluation and Selection
- Techniques to Improve Classification Accuracy: Ensemble Methods

Model Evaluation and Selection

- Evaluation metrics: How can we measure accuracy? Other metrics to consider?
- Use **validation test set** of class-labeled tuples instead of training set when assessing accuracy
- Methods for estimating a classifier's accuracy:
  - random subsampling
  - k-fold, leave one out and stratified cross-validation
  - Bootstrap
- Comparing classifiers:
  - Confidence intervals
  - Cost-benefit analysis and ROC Curves

Classifier Evaluation Metrics: Confusion Matrix

Confusion Matrix:

Actual class\Predicted class	C_1	$\neg C_1$
C_1	**True Positives (TP)**	**False Negatives (FN)**
$\neg C_1$	**False Positives (FP)**	**True Negatives (TN)**

Example of Confusion Matrix:

Actual class\Predicted class	buy_computer = yes	buy_computer = no	Total
buy_computer = yes	**6954**	**46**	7000
buy_computer = no	**412**	**2588**	3000
Total	7366	2634	10000

- Given *m* classes, an entry, $\boldsymbol{CM_{i,j}}$ in a **confusion matrix** indicates # of tuples in class *i* that were labeled by the classifier as class *j*
- May have extra rows/columns to provide totals

Classifier Evaluation Metrics: Accuracy, Error Rate, Sensitivity and Specificity

A\P	C	¬C	
C	**TP**	**FN**	**P**
¬C	**FP**	**TN**	**N**
	P'	**N'**	**All**

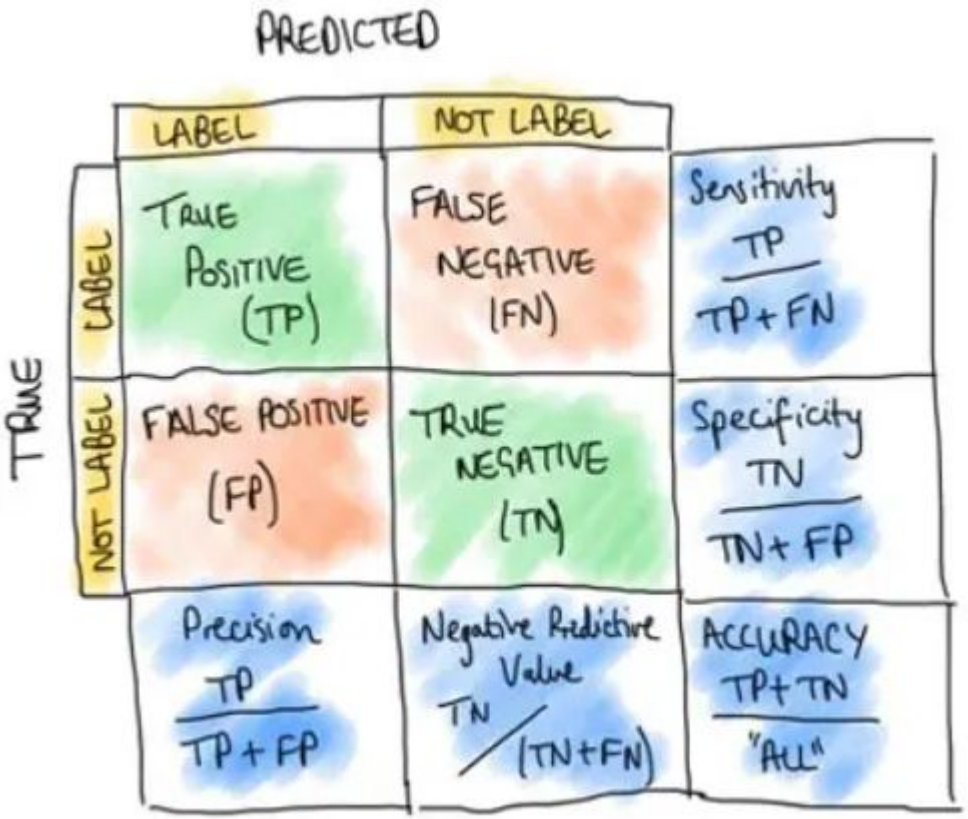
- **Classifier Accuracy,** or recognition rate: percentage of test set tuples that are correctly classified

  Accuracy = (TP + TN)/All

- **Error rate:** *1 – accuracy*, or

  Error rate = (FP + FN)/All

- **Class Imbalance Problem**:
  - One class may be *rare*, e.g. fraud, or HIV-positive
  - Significant *majority of the negative class* and minority of the positive class
  - **Sensitivity**: True Positive recognition rate
    - **Sensitivity = TP/P**
  - **Specificity**: True Negative recognition rate
    - **Specificity = TN/N**


PREDICTED
LABEL
NOT LABEL
TRUE
LABEL
NOT LABEL
TRUE POSITIVE (TP)
FALSE NEGATIVE (FN)
Sensitivity TP / TP+FN
FALSE POSITIVE (FP)
TRUE NEGATIVE (TN)
Specificity TN / TN+FP
Precision TP / TP+FP
Negative Predictive Value TN / (TN+FN)
ACCURACY TP+TN / "ALL"

Classifier Evaluation Metrics: Precision and Recall, and F-measures

- **Precision**: exactness—what percentage of tuples that the classifier labeled as positive are actually positive?

$$precision = \frac{TP}{TP + FP}$$

- **Recall:** completeness – what % of positive tuples did the classifier label as positive?

$$recall = \frac{TP}{TP + FN}$$

- Perfect score is 1.0
- Inverse relationship between precision & recall
- ***F* measure (*F*$_1$ or *F*-score)**: harmonic mean of precision and recall,

$$F = \frac{2 \times precision \times recall}{precision + recall}$$

Classifier Evaluation Metrics: Example

Actual Class\Predicted class	cancer = yes	cancer = no	Total	Recognition(%)
cancer = yes	**90**	**210**	300	30.00 (*sensitivity*
cancer = no	**140**	**9560**	9700	98.56 (*specificity)*
Total	230	9770	10000	96.40 (*accuracy*)

- *Precision* = 90/230 = 39.13% *Recall* = 90/300 = 30.00%

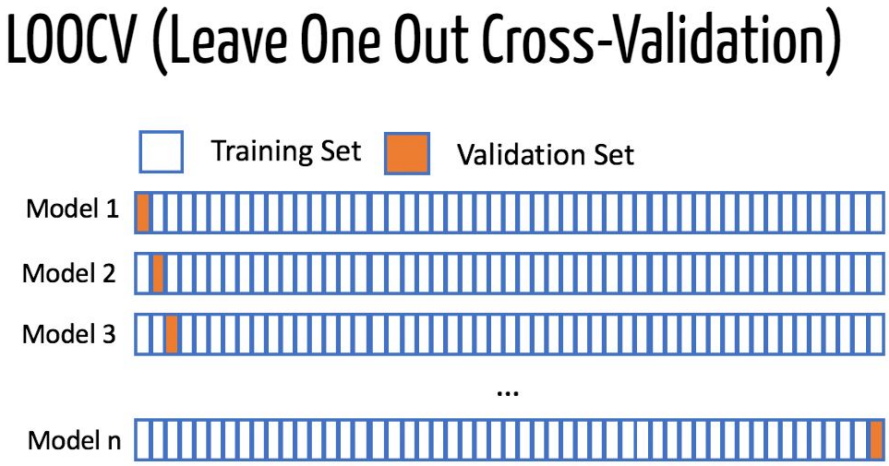
Evaluating Classifier Accuracy: Holdout & Cross-Validation Methods

- **Holdout method**
  - Given data is randomly partitioned into two independent sets
    - Training set (e.g., 2/3) for model construction
    - Test set (e.g., 1/3) for accuracy estimation

  - <u>Random sampling</u>: a variation of holdout
    - Repeat holdout k times; accuracy = avg. of the accuracies obtained

Evaluating Classifier Accuracy: Holdout & Cross-Validation Methods

- **Cross-validation** (k-fold, where k = 10 is most popular)
  - Randomly partition the data into *k mutually exclusive* subsets, each approximately equal size
  - At *i-th iteration,* use D_i as test set and others as training set
  - <u>Leave-one-out</u>: *k* folds, where *k* = # of tuples, for small-sized data
  - Stratified cross-validation: folds are stratified so that class dist. in each fold is approximately the same as that in the initial data (link)



Evaluating Classifier Accuracy: Bootstrap

- **Bootstrap**

  Works well with small data sets

  Samples the given training tuples uniformly *with replacement*

  i.e., each time a tuple is selected, it is equally likely to be selected again and re-added to the training set

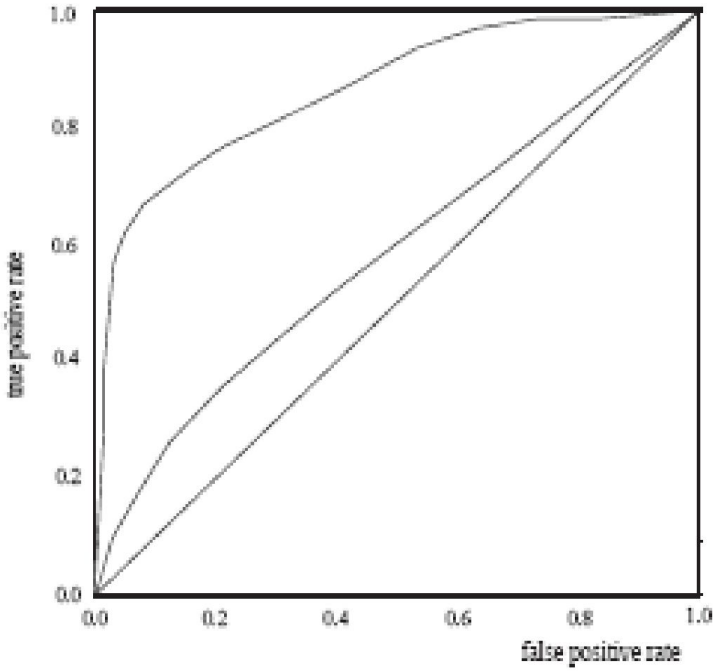
  A data set with *d* tuples is sampled *d* times with replacement, resulting in a training set of *d* samples. The data tuples that did not make it into the training set end up forming the test set. Repeat the sampling procedure *k* times.

home work: compare bootstrap and cross validation

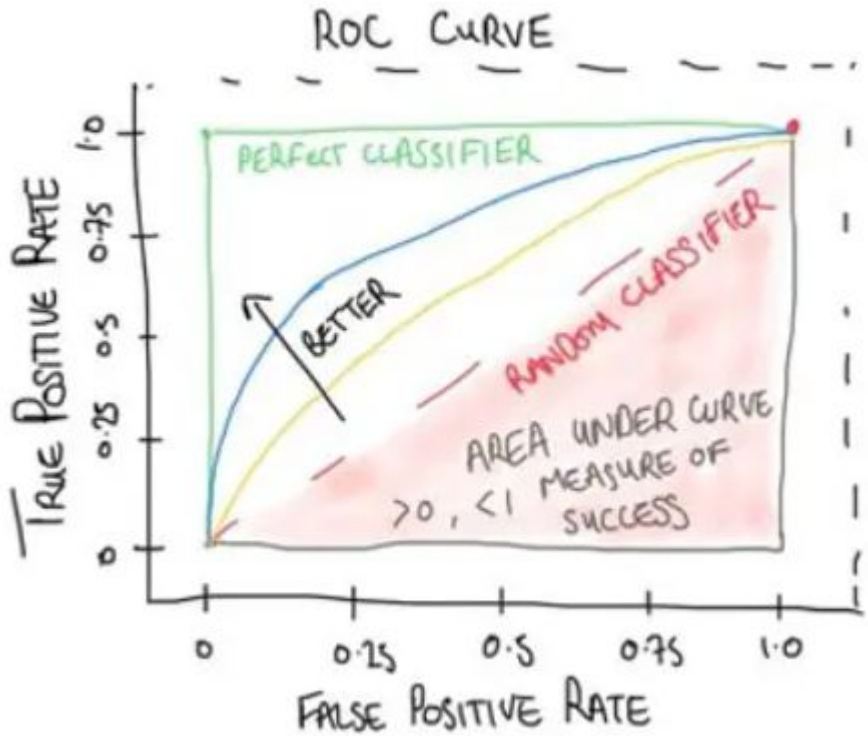
	cross validation	bootstrap
advantages:		
disadvantages:		
when to use:		
good to use for what:		

Model Selection: ROC Curves

- **ROC** (Receiver Operating Characteristics) curves: for visual comparison of classification models
- Originated from signal detection theory
- Shows the trade-off between the true positive rate and the false positive rate
- The area under the ROC curve is a measure of the accuracy of the model
- Rank the test tuples in decreasing order: the one that is most likely to belong to the positive class appears at the top of the list
- The closer to the diagonal line (i.e., the closer the area is to 0.5), the less accurate is the model



- Vertical axis represents the true positive rate
- Horizontal axis rep. the false positive rate
- A model with perfect accuracy will have an area of 1.0



https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database/download?datasetVersionNumber=1

Predictor Error Measures

- Measure predictor accuracy: measure how far off the predicted value is from the actual known value
- **Loss function**: measures the error between y_i and the predicted value y_i'
  - Absolute error: $| y_i - y_i' |$
  - Squared error: $(y_i - y_i')^2$
- Test error (generalization error): the average loss over the test set
  - Mean absolute error: $\frac{\sum_{i=1}^{d} | y_i - y_i' |}{d}$ Mean squared error: $\frac{\sum_{i=1}^{d} (y_i - y_i')^2}{d}$
  - Relative absolute error: $\frac{\sum_{i=1}^{d} | y_i - y_i' |}{\sum_{i=1}^{d} | y_i - \bar{y} |}$ Relative squared error: $\frac{\sum_{i=1}^{d} (y_i - y_i')^2}{\sum_{i=1}^{d} (y_i - \bar{y})^2}$

The mean squared-error exaggerates the presence of outliers

Popularly use (square) root mean-square error; similarly, root relative squared error